EXERCISES ABOUT CHAPTER 10

Gases

- For a gas sample at 25.6°C and 748 mmHg; its volume 4.25 L.
  What is the volume of this gas sample at 26.8°C and 742 mmHg?

$$PV = nRT$$

$$nR = \frac{PV}{T}$$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \longrightarrow \frac{748\ mmHg \times 4.25\ L}{(25.6+273)\ K} = \frac{742\ mmHg \times V_2}{(26.8+273)\ K}$$

$$V_2 = 4.30\ L$$

- At 46°C, 35.8 g O_2 gas exists in a cylinder with 12.8 L volume. Calculate the pressure in atm.

$$P = \frac{n \cdot R \cdot T}{V} = \frac{\left(\frac{35.8\ g\ O_2}{32\ g/mol}\right) \times 0.082 \frac{L \cdot atm}{mol \cdot K} \times (46 + 273)K}{12.8\ L}$$

$$P = 2.29\ atm$$

- Ammonia synthesis reaction is given

$$N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$$

Consider all amount of initial gases has been reacted and gases are ideal gas.

a) At 315°C and 5.25 atm, what would be the volume of product $NH_{3(g)}$ from 313 L H_2 ?
b) Calculate the volume of the product $NH_{3(g)}$ at 25°C, 727 mmHg.

a) 313 L H_2 ⟶ $$n = \frac{PV}{RT} = \frac{5.25\ atm\ \times 313\ L}{0.082\ \frac{L \cdot atm}{mol \cdot\ K} \times 588K} = 34.08\ mol$$

$$N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$$

11.36 mol 34.08 mol ⟶ 22.72 mol

$$V = \frac{nRT}{P} = \frac{22.72\ mol \times 0.082\ \frac{L \cdot atm}{mol \cdot K} \times 588K}{5.25\ atm}$$

$$V = 208.66\ L$$

b)

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\frac{5.25\ atm \times 208.66\ L}{588\ K} = \frac{0.956\ atm\ \times V_2}{298\ K}$$

$$V_2 = 580.74\ L$$

- 2 L vessel is filled with $Ar_{(g)}$ at 752 mmHg and 35°C. Then 0.728 g C_6H_6 vapor is added to this vessel.

a) What is the total pressure in the vessel?
b) What are the partial pressures of each gas?

$$P_{C_6H_6} = \frac{nRT}{V} = \frac{\left(\frac{0.728\ g}{78.11\ g/mol}\right) \times 0.082\ \frac{L \cdot atm}{mol \cdot K} \times 308\ K}{2\ L} \times \frac{760\ mmHg}{1\ atm} = 89.5\ mmHg$$

$$P_{total} = 752\ mmHg + 89.5\ mmHg = 842\ \text{mmHg}$$

- A mixture of 6.0 g $O_{2(g)}$ and 9.0 g $CH_{4(g)}$ is placed in a 15 L vessel at 0°C.

What is the partial pressure of each gas and total pressure?

$n_{O_2} = 0.1875\ mol$

$n_{CH_4} = 0.5625\ mol$

At 0°C
V = 15 L

$P_{O_2} \cdot 15\ L = n_{O_2} \cdot R \cdot T$

$P_{O_2} \cdot 15\ L = 0.1875\ mol \cdot 0.082\ \frac{L \cdot atm}{mol \cdot K} \cdot 273\ K \longrightarrow P_{O_2} = 0.28$ atm

$P_{CH_4} \cdot 15\ L = 0.5625\ mol\ \cdot 0.082\ \frac{L \cdot atm}{mol \cdot K} \cdot 273\ K \longrightarrow P_{CH_4} = 0.84\ atm$

$P_{total} = 1.12\ atm$

- In a specific time range 0.00484 moles $N_2O_{(g)}$ effused through a pinhole. At the same conditions and same time how much does $NO_{2(g)}$ effuse?

$$\frac{r_1}{r_2} = \frac{u_{\text{rms1}}}{u_{\text{rms2}}} = \sqrt{\frac{3RT/\mathcal{M}_1}{3RT/\mathcal{M}_2}} = \sqrt{\frac{\mathcal{M}_2}{\mathcal{M}_1}}$$

$$\frac{r_{NO_2}}{r_{N_2O}} = \sqrt{\frac{M_{N_2O}}{M_{NO_2}}} = \sqrt{\frac{44.02}{46.01}} = 0.9781 = \frac{x\ mol\ NO_2}{0.00484\ mol\ N_2O} \longrightarrow$$ 0.00473 mol NO_2